

AI Automated Employee Onboarding System

1. Problem definition & scope -15p

Clear business/IT problem, objective, scope, assumptions, exclusions.

Problems:

- Onboardingul este un proces de lunga durata
- De multe ori se realizeaza incomplet
- Pierdere de timp si pentru angajat, si pentru echipele de IT/HR

Objective:

- Eficientizarea procesului prin automatizarea partiala cu AI
- Minimizarea necesitatii de interventie umana

Scope:

- Crearea unei platforme care unifica procesele de onboarding si offboarding
- Reducerea timpului folosit de echipele de HR in procese

Assumptions:

- Datele primite sunt procesate de HR
- Produsele SaaS functioneaza pe modelul seats/entitlements

Exclusions:

- Date ale angajatului care nu vizeaza atribuirea de accese/licente

2. Understanding of the process - 15p

Metoda traditionala, blocaje, ce se poate imbunatati cu AI

Traditional:

HR Email/Ticket -> IT Helpdesk -> Triage -> SysAdmin L2 Slack/Email -> Security



1. HR trimite un email sau deschide un tichet Jira/ServiceNow in legatura cu angajatul nou la Finance
2. Tehnicianul citeste tichetul si observa lipsa unor detalii specifice pt. angajat
3. IT discuta cu Finance Manager pe Slack despre ce licente sa aloce angajatului; ore mai tarziu, primeste un raspuns. (IT + Manager, 2 actiuni)
4. Tehnicianul creeaza cont + parola in AD/Google WS cu o parola temporara si credentiale trimise, probabil, nesigur, pe o adresa personala de mail a angajatului
5. Tehnicianul cauta si gaseste in depozit un laptop pentru noul angajat;
6. Se aloca licentele SaaS de catre tehnician: 365 Admin Center, PowerBI Service, JetBrains, Jira;
7. Exista posibilitatea unei lipse de comunicare intre care rezulta intr-un timp mai mare de asteptare, deoarece angajatul poate sa nu aibe, for example, acces la VPN-ul Finance. Wait time, inca 24h.

Numar de actiuni umane: minim 7 interventii manual, fragmentate, si predispuse erorilor

- erori la acordare de permisiuni → timp pierdut
- intarzieri operationale
- politici nestructurate, greu de urmarit

- time efficient
- poate urmări politici mai bine
- traceable

Cum arata sistemul care va imbunatati problema

care generează plan complet aprobare HITL

- se extrage setul default pentru Data Analyst (Google WS, PowerBI Pro, DataGrip, Laptop Dell XPS 15)
- verificare RAG a policy-ului intern, gaseste clauza, marcheaza flagul
- se asambleaza planul complet si bine structurat

4. Se apeleaza tool-urile deterministice: `assign_license()`, `assign_training()`

Time reduction de >95% a timpului operational si zero day-1 blockers.

Diagrame tehnice si workflow. Components and relationships: input, model/LLM, data, retrieval if relevant, application/backend, UI, external systems, output.



2. ORCHESTRATION & REASONING (FastAPI + LLM Agent)

Context +

Deterministic Lookups

3A. STRUCTURED DATA (SQL)
(Supabase)
-Employees & Roles Matrix
-Software Products & Seats
-Inventory / IT Asset Records

3B. VECTOR RETRIEVAL (RAG)
(ChromaDB)
-Security Access Policies
- Compliance & Retention Guidelines
- Exception & Escalation Rules

4. HUMAN-IN-THE-LOOP (HITL) GATEWAY
IT Admin inspects plan, policy citations, & diffs
APPROVE / REJECT

(Upon approval)

5. DETERMINISTIC TOOL EXECUTION ENGINE
- provision_account(email, role)|
- assign_license(emp_id, prod_id)
- allocate_asset(emp_id, asset_type)
- revoke_license(emp_id, prod_id)

6. AUDIT, STATE PERSISTENCE & OUTPUT
-Update seats, create ENT-IDs, write Audit Log
-Generate Temporary Access Pass (TAP) activation link

5. Data design & RAG thinking -15p (pentru cine e aplicabil)

What data is required, approximate entities/schema, mock data strategy, what information needs retrieval/search, and possible ChromaDB usage.

Input data structure example:

```
{
  "request_id": "ONB-104",
  "employee_id": "EMP-204",
  "first_name": "Ana",
  "last_name": "Pop",
  "role": "Data Analyst",
  "department": "Finance",
  "manager_id": "EMP-050",
  "location": "Romania",
  "start_date": "2026-09-01",
  "employment_type": "full_time",
  "hr_approved": true
}
```

Database table schemas:

software_products

- product_id
- name
- license_type
- total_seats
- requires_approval

role_entitlement_rules

- role_id
- product_id
- access_level
- mandatory
- approval_required

license_assignments

- employee_id
- product_id
- status
- assigned_at
- revoked_at

functii mock de license assignment

information that needs retrieval/search:

what licenses a post needs(Data Analyst → Microsoft Office, JetBrains)

RAG/ChromaDB usage:

storing/retrieving policy clauses, providing explanations

6. Reasoning / decision / execution concept -10p

Identificare a agentilor (nu foarte explicit pentru ca urmeaza sa discutam in modulul urmator)

Onboarding Agent that proposes actions in JSON/Pydantic format

7. KPIs & success criteria -10p

1–2 sensible measures showing whether the project works better than the current process. doar trebuie sa alegeti o metoda prin care veti putea verifica rata de succes, nu e nevoie deja de estimare sau demo

a) Average time to onboard new employee

b) Average fail rate